Rohit Menon

Robotics Researcher | Perception, Reasoning and Manipulation

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Website | Google Scholar | GitHub | LinkedIn



Profile

- Robotics researcher in **deep learning, scene graphs, active perception and mobile manipulation**
- Former **Neura Robotics founding engineer and AI group leader**; built MAiRA's perception–manipulation and hard real-time control systems

Professional Experience

Research Associate (Wissenschaftlicher Mitarbeiter) Feb. 2022–Present
Humanoid Robots Lab, University of Bonn, Germany

- Delivered active-perception and manipulation systems across **PhenoRob, Lamarr and AID4Crops**; authored six first- or equal-first-author papers.
- Managing procurement of robotics hardware exceeding **EUR 100,000** and directly supervising **three student assistants**.
- Mentoring three junior PhD students; supervised five MSc theses, yielding **three papers and two thesis awards**.
- Contributed to the **Agriloop proposal (EUR 400,000)** on foundation models for human-in-the-loop greenhouse robotics.
- Presented robotics through a **WDR Bonn interview**, **ICRA@40** and **public demonstrations** in Bonn.

PhenoRob
  - Developed greenhouse-tested fruit mapping (NBV-SC) and a three-arm harvesting system (HortiBot; Dawn-IK).
  - Developed uncertainty-aware depth and semantic mapping, evaluated on NYUv2 and zero-shot ScanNet (EvidMTL/EvidKimera).

Lamarr
  - Developed open-vocabulary object search using LLM priors, scene geometry and probabilistic scene graphs (O-STaR).

AID4Crops
  - Developed graph reasoning for plant-part relations and directional occlusions; first author, IROS 2026 (SG-DOR).

Founding Engineer / Group Leader, AI for Manipulation Nov. 2019–Jan. 2022
Neura Robotics GmbH, Metzingen, Germany

- **Built MAiRA's perception–manipulation architecture** as the first AI engineer, including C++/ROS drivers and interfaces
- Re-engineered Xenomai/Orocos RTT control tasks over SOEM EtherCAT, improving **joint-tracking accuracy by 10×**
- Developed investor-facing robot demonstrations for the company's **2020 pre-Series-A fundraising effort**
- Developed vision-based pick-and-place with **voice control** and built a **robot tic-tac-toe** demonstration.
- Built RGB, depth and radar fusion for real-time **human detection and collision avoidance**.
- Was promoted to lead **AI for Manipulation** with hiring responsibility; presented MAiRA at **GITEX 2021** (media).

Researcher, Robot Manipulation and Control July 2015–Oct. 2019
DFKI Robotics Innovation Centre, Bremen, Germany

- Developed **planning, control, teleoperation and manipulation software** for high-DoF, mobile and humanoid robots.
- Led DFKI's iLEVATOR contribution and managed approximately **EUR 100,000** in project finances, procurement and reporting.

NeTTUN
  - Owned the control and communication architecture for a tunnel-boring manipulator, including jerk-limited trajectories and teleoperation.

iLAADR / iLEVATOR
  - Built monocular known-object picking from CAD templates; deployed iLEVATOR bin handling at the end-user factory.

Hybr-iT / TransFit
  - Developed mobile-manipulator planning, unknown-object picking, reactive grasping and RH5 arm planning.

MSc Thesis Student, Shared Autonomy June 2014–Feb. 2015
DLR Institute of Robotics and Mechatronics, Oberpfaffenhofen, Germany

- Developed **adaptive shared autonomy** using automatic grasp selection and virtual fixtures while preserving approach control
- Validated the method in **full-dynamics simulation, a user study and on the real robot**; published at ICRA 2016

Engineering and Commissioning / Senior Executive Sept. 2008–Sept. 2012
Siemens Ltd, Thane, India

- Developed **drive controls and PLC/HMI software**; commissioned machinery and was **promoted to Senior Executive within two years**.
- Supported the sales team in winning multiple automation projects totaling more than **EUR 1 million**.

Education

PhD in Computer Science Feb. 2022–Mar. 2027 (Expected)
University of Bonn, Germany | Advisor: Prof. Dr. Maren Bennewitz

MSc in Automation and Robotics Oct. 2012–Mar. 2015
TU Dortmund University, Germany | Grade: 1.2; top 5% of cohort

Bachelor's in Electrical Engineering Aug. 2004–June 2008
SPCE, University of Mumbai, India | Grade: 76%; silver medal

Skills

Research: Deep learning; graph neural networks; shape completion; scene graphs; uncertainty quantification; 3D semantic mapping; active perception; open-vocabulary mobile manipulation
Programming and learning: C++, Python, MATLAB, PyTorch, PyTorch Geometric
Robotics and simulation: ROS, ROS 2, Isaac Sim, Gazebo, Blender
Control and libraries: Xenomai, Orocos RTT, SOEM EtherCAT, OpenCV, PCL, Eigen, NLopt
Languages: English C2 (TOEFL: 116/120), German B2, Hindi, Malayalam

Selected Awards

Best Poster Award, IROS 2025 Workshop on Perception and Planning for Mobile Manipulation in Changing Environments.
Scholarships: Deutschlandstipendium (2013), Ratan Tata scholarship (2006, 2007).
Industry award: Siemens Excellence in Commissioning trophy (2011).

Selected Publications

1. **Rohit Menon***, Niklas Mueller-Goldingen*, Sicong Pan, Gokul Krishna Chenchani, Maren Bennewitz. "SG-DOR: Learning Scene Graphs with Direction-Conditioned Occlusion Reasoning for Pepper Plants." *IROS*, 2026
2. **Rohit Menon**, Yasmin Schmiede, Gokul Krishna Chenchani, Hermann Blum, Maren Bennewitz. "O-STaR: Effective Object Search through Spatio-Temporal Reasoning on Dynamic Scene Graphs." *GRC*, 2026; *ICRA*, 2027, under review
3. **Rohit Menon**, Nils Dengler, Sicong Pan, Gokul Krishna Chenchani, Maren Bennewitz. "EvidMTL: Evidential Multi-Task Learning for Uncertainty-Aware Semantic Surface Mapping." *IROS*, 2025
4. Christian Lenz*, **Rohit Menon***, Michael Schreiber, Melvin Paul Jacob, Sven Behnke, Maren Bennewitz. "HortiBot: An Adaptive Multi-Arm System for Robotic Horticulture of Sweet Peppers." *IROS*, 2024
5. Salih Marangoz*, **Rohit Menon***, Nils Dengler, Maren Bennewitz. "DawnIK: Decentralized Collision-Aware Inverse Kinematics for Multi-Arm Systems." *Humanoids*, 2023
6. **Rohit Menon**, Tobias Zaenker, Nils Dengler, Maren Bennewitz. "NBV-SC: Shape-Completion-Based Next-Best-View Planning for Fruit Mapping." *IROS*, 2023
7. Jörn Vogel, Katharina Hertkorn, **Rohit U. Menon**, Máximo A. Roa. "Flexible, Semi-Autonomous Grasping for Assistive Robotics." *ICRA*, 2016

**Equal contribution.*

References

Prof. Dr. Maren Bennewitz
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Dr. Milad Malekzadeh
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